

Progga Paromita Dutta

(352) 792 3852 • Stony Brook, NY • progga.buetme18@gmail.com • [LinkedIn](#) • [GitHub](#) • progga004.github.io

EDUCATION

Stony Brook University – Stony Brook, NY

Jan 2022 - Dec 2024

Bachelor of Science with Honors: Computer Science | Applied Mathematics and Statistics

GPA: 3.67/4.0

Recipient of Dean's list Award, 2022 (Spring, Fall) -2024 (Spring)

Related Coursework: Software Development I & II, Analysis of Algorithm, Machine learning, Computer Vision, System Fundamentals I & II, Theory of Computation, Data Structures, Object-Oriented Programming

Bangladesh University of Engineering & Technology – Dhaka, Bangladesh

Jun 2019 - Dec 2021

Bachelor of Science in Mechanical Engineering

GPA: 3.77/4.0

SKILLS

- **Programming Languages:** Python (Numpy, Pandas, OpenCv, TensorFlow, Pytorch, Scikit-learn, Matplotlib), HTML/CSS, JavaScript, SQL, React, Node.js, Java, C, R, SAS, Matlab, Ocaml, Assembly
- **Technologies:** Git, GitHub, Bitbucket, Eclipse, Visual Studio, LaTeX, MongoDB, CircleCI, Microsoft Office
- **Activities:** Member- Society of Women in Engineering (SWE), Stony Brook Ballroom Dance Team

RESEARCH EXPERIENCE

Yin Lab, Stony Brook University

Stony Brook, NY

Research Assistant, Supervisor: Dr. Zhaozheng Yin

Jan 2024 – Present

- Driving independent research to enhance eye gaze estimation accuracy by leveraging 468 facial landmarks, building on existing eye gaze estimation models.
- Developing a communication board for individuals with cerebral palsy that combines eye-tracking with facial feature analysis, using data from both eye regions and facial landmarks for improved interaction.

Computer Vision Lab, Stony Brook University

Stony Brook, NY

Research Assistant, Human Computer Interaction Lab

May 2023 – May 2024

- Collaborated with a team of 6 people on a human gaze target prediction project using multi-camera setups, contributing to data collection and accurate labeling with VGG Image Annotator.
- Improved expertise in project methodologies and terminology, focusing on data annotation, calibration, and multi-view integration to support future work in activity recognition, social behavior analysis, and human-computer interaction.

PROFESSIONAL EXPERIENCE

Evolv Technology

Waltham, MA

Computer Vision Engineer Intern, Advanced Threat Detection (ATD) Team

June 2024- August 2024

- Designed and integrated a custom tracker in Evolv's weapon detection system using Kalman filter and centroid-based tracking algorithm, improving object identity consistency across video frames.
- Optimized a script to efficiently process and store 3.2 million images locally, reducing processing time from 3 weeks to 32 minutes through profiling techniques, providing essential data for training custom object detectors with models like Grounding DINO.

TEACHING EXPERIENCE

Computer Science Department, Stony Brook University

Stony Brook, NY

Teaching Assistant, Foundations of Computer Science

August 2022 - December 2022

- Mentored students on functions, relations, sets, recursion, logic, mathematical induction, proof techniques through office hours and weekly problem-solving sessions for groups of 30 students.
- Assisted with classroom management and oversight during examinations in collaboration with the professor.

Applied Mathematics & Statistics Department, Stony Brook University

Stony Brook, NY

Teaching Assistant, Finite Mathematical Structures

August 2022 - December 2022

- Instructed approximately 180 students in a Finite Mathematical Structures course through weekly two-hour office hours, covering topics like graph theory, combinatorics, and recurrence relations.
- Aided the professor with exam grading, classroom logistics, and supervision to ensure efficient course operations and student engagement.

LEADERSHIP & SERVICE

Undergraduate Admissions Office, Stony Brook University

Admissions Tour Guide

Stony Brook, NY

May 2023 - Present

- Lead and manage tours with up to 120 visitors, ensuring a seamless and engaging experience across SBU Campus locations.
- Deliver informative tours to diverse groups, demonstrating exceptional communication and public speaking skills to educate and captivate visitors.

Undergraduate Student Government, Stony Brook University

Voting Member, Senate Committee

Stony Brook, NY

August 2023- Spring 2024

- Engaged in policy formulation and decision-making processes to advocate for student interests and enhance the university environment, representing a diverse student body in the senate.
- Collaborated on the planning and execution of campus initiatives, managing resources and organizing events to foster community engagement and improve student life.

International Youth Summit, WSDA, New Zealand

County Manager

Dhaka, Bangladesh

May 2021- Jan 2022

- Led a team of approximately 400 people as the main leader by forming small groups and assigning them weekly works to complete and supervising them weekly.
- Introduced the project for the well-being of the youth generation to make themselves worthy as an individual and for their career development.

PUBLICATIONS

Qiaomu Miao, Vivek Raju Golani, **Progga Paromita Dutta**, Jingyi Xu, Minh Hoai, Dimitris Samaras(November 2024). Multi-view Gaze Target Estimation. Submitted to CVPR Conference 2025(under review)

AWARDS

- SBU Far Beyond Scholarship – Awarded annually based on academic achievement; \$3000 per year
- Design Automation Conference Young Fellow Scholarship (2023)
- Society of Women Engineers (SWE) Conference Attendee 2024

PROJECTS

- **Communication Board Development - Full Face Appearance Based Eye Gaze Estimation| Python, MediaPipe**
Conducting independent research to develop a communication board for cerebral palsy people using facial feature extraction for eye gaze estimation, implementing advanced machine learning models for real-time interaction.
- **Custom Object Tracker Development - Con-Op Compliance Verification | Python, RT-DETR**
Implemented a custom object tracker integrated with RT-DETR to detect and track laptops and persons in real-time, developing an algorithm to assess Con-Op compliance by monitoring relationships between detected objects while maintaining detection accuracy and reliability.
- **The Hospital Project - Process Management Tool | React, MongoDB, Node.js, Express**
Developed a tool to enhance departmental efficiency by managing accounts, resources, equipment, procedures, and staff assignments, enabling dynamic decision-making and caregiver notifications.
- **Fake Stack Overflow Application | React, MongoDB, Node.js, Express, bcrypt**
Engineered a Stack Overflow-inspired app using React, Node.js/Express, and MongoDB, implementing secure authentication with bcrypt and translating user stories into a responsive UI and efficient server-side routing.
- **Homography Estimation and image Warping| Python, OpenCv**
Implemented advanced computer vision techniques for image alignment and stitching producing a seamless image mosaic that simulates panoramic photography using feature matching algorithms and Random Sample Consensus (RANSAC) for robust homography estimation.
- **Neural Network Framework| Python, Numpy**
Developed a versatile neural network framework, enabling model training for classification and regression with dynamic architectures activation functions, and optimized loss computations.